

THE UNIVERSITY OF QUEENSLAND
SCHOOL OF CIVIL ENGINEERING
CIVL4333 Advanced Concrete Design
Assignment 4

Part A)

A bridge section (Figure 1) is subject to combined factored design loads of T^* , M^* and V^* . Determine the longitudinal and transverse reinforcement required, using the sectional design approach in AS3600.

Relevant details are as follows:

- $T^* = (x/3)$ kNm; $V^* = (4x/3)$ kN; $M^* = 800$ kNm - where x is the first 3 digits of **your** 8-digit student number. For example: If **your** 8-digit student number is **412**34567, your T^* is 137.3 kNm and V^* is 549.3 kN.
- $f'_c = 40$ MPa; $f_{sy} = f_{sy.f} = 500$ MPa.
- Clear cover to reinforcement = 30 mm.
- Assuming N12 stirrups and N32 longitudinal bars.
- Using relevant clauses in AS3600-2018.

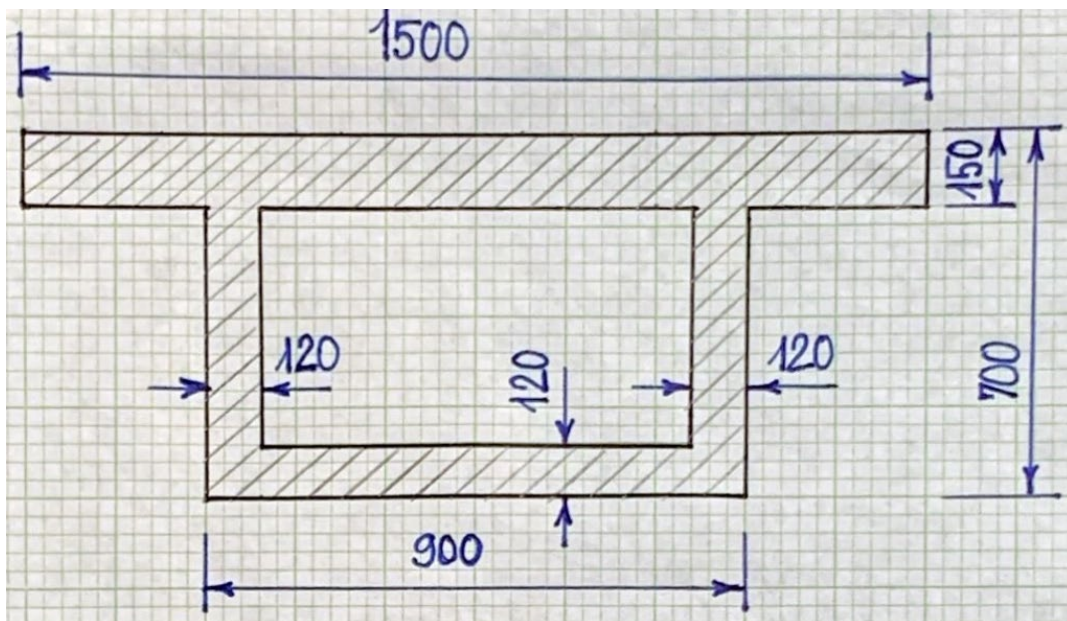


Figure 1.

Part B)

The section in Question 1 is now also subject to a uniform pre-compression. Comment on how each of the following would change due to the presence of such uniform pre-compression: (i) the cracking torque; (ii) the angle of inclination between the initial cracks and the longitudinal axis; and (iii) the required longitudinal and transverse reinforcement.

END OF ASSIGNMENT 4